

Effects of Gamification and Physiological Stress on Webstore Retention and Purchase Intentions

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Abstract

Gamification features have been found to positively affect engagement, enjoyment, and purchase behavior in webstore marketplaces. While gamified environments can simultaneously cause cognitive load and stress, the interplay between gamification effects and stress is not well understood. Moreover, there are no studies considering objective measures such as physiological heart rate—calculated stress in the context of gamification and e-commerce research. Our research consists of two studies. The first analyzes the effects of perceived gamification and stress on webstore retention, and the second assesses these effects on purchase intentions. For physiological stress measurement, an arm strap sensor (electrical activity of the heart, ECG) was used together with a visit to a selected webstore and survey questions. Our findings offer important implications for webstore designers and marketers developing gamification. First, these practitioners should reduce customer stress through education so that customers feel less stressed when making e-commerce selections. Moreover, while incorporating gamification features, they should avoid increasing stress. Specifically, we recommend avoiding challenging environments, reward systems, and game-like experiences for stressed consumers, while enabling feelings of achievement instead. This study opens a new avenue in stress and gamification research, calling for future work to validate these results with further experiments and contexts.

CCS Concepts

• **mpirical studies in HCI**; • **HCI design and evaluation methods**; • **Electronic commerce**; • **Health informatics**;

Keywords

Gamification, E-commerce, Heart rate variability, Physiological stress

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1 Introduction

Gamification features have been found to have positive effects in enhancing engagement, enjoyment, and purchase behavior in webstore marketplaces [1, 2]. While gamified environments can simultaneously cause cognitive load and stress [3], the interplay of gamification effects and stress is not well understood. Moreover, existing gamification and e-commerce research has largely overlooked objective indicators of stress, such as physiological measures derived from heart rate data. This omission is notable given that physiological measures have been shown to be more accurate predictors of behavior than self-reported psychological measures alone [4, 5]. In addition, the increasing adoption of physiological sensing technologies among consumers presents new opportunities for marketers and researchers alike [6].

2 Background literature: Gamification and e-commerce research

Gamification is defined as a design approach that aims to create experiences similar to those found in games, with the goal of influencing user behavior and cognitive processes to benefit marketing outcomes [7]. Central to designing gamification is the inclusion of features and elements such as badges, points, levels, leaderboards, and progress indicators, as well as immersion, narrative, and social components. As a result, gamification is not just about adding game elements, but about designing game-like experiences that engage users emotionally, cognitively, and socially [7]. Gamification has been found to positively enhance engagement, enjoyment, and purchase intention and behavior in e-commerce marketplaces [1, 2]. This enhancement occurs especially through satisfying utilitarian, social, and experiential value [7, 8], motivations [9], as well as aesthetics, immersion [10, 11], and flow [12]. Moreover, especially game-task modularity and self-efficacy have been found to enhance these effects [13]. Gamification has also been found to positively affect repurchase intentions and customer retention in webstores [14]. While gamified environments can simultaneously cause cognitive load and stress, as recognized by Turan et al. (2016) in educational settings, the effect and interplay of stress with gamification have not yet been addressed in the marketing literature.

3 Heart rate variability and stress measures in marketing research

Stress is a state in which the body's homeodynamic balance is threatened by a wide range of intrinsic or extrinsic, real or perceived challenges—collectively known as stressors [15, 16]. Homeodynamic balance is affected by all bodily systems, including nervous, hormonal, immune, cardiovascular, and metabolic systems [16]. To maintain homeodynamic balance, the body has a highly

sophisticated stress-regulation system, characterized by its inherent capacity for self-regulation and adaptability [17–19]. One of these self-regulation systems is *emotional regulation*, which plays a crucial role in how individuals experience and manage stress [20, 21]. Effective emotion regulation strategies, such as cognitive reappraisal and intentional exploration of emotions, are associated with better adjustment, lower perceived daily stress, and greater well-being. On the other hand, maladaptive strategies like emotion suppression can increase negative stress effects [22].

As noted above, the nervous system is one of the bodily systems affecting homeodynamic balance. The nervous system is further divided into the somatic and autonomic nervous systems, the former operating under conscious control and the latter under unconscious control [23]. The *autonomic nervous system (ANS)* comprises two main branches: the sympathetic nervous system (SNS), commonly referred to as the “fight or flight” division that prepares the body for action and the parasympathetic nervous system (PNS), often described as the “rest and digest” division that promotes relaxation and recovery [24].

Today, wearable technologies enable continuous, non-intrusive monitoring of both emotional regulation and ANS through measures such as heart rate variability (HRV) [25]. For example, the ratio of SD2 (standard deviation of long-term variability) to SD1 (standard deviation of instantaneous beat-to-beat variability), collectively referred to as SD2/SD1, as measured by HRV, indicates autonomic nervous system (ANS) balance and emotional regulation [26, 27]. Specifically, SD1 represents short-term, beat-to-beat variability in heart rate reflecting PNS activity. SD2 represents longer-term variability in heart rate influenced by both SNS and PNS activity, though sympathetic activity has a stronger contribution. As a result, the SD2/SD1 ratio reflects the balance between long-term (SNS) and short-term (PNS) influences. A higher SD2/SD1 ratio may indicate sympathetic “fight or flight” dominance, while a lower ratio suggests parasympathetic “rest and digest” predominance. By combining both SNS and PNS influences, the SD2/SD1 ratio is a good indicator of long-term stress [28].

Short-term parasympathetic activity linked to relaxation, concentration, and emotional control is measured by another metric called RMSSD (Root Mean Square of Successive Differences), which results in increased short-term stress. [29–33].

Stress measures in marketing research have long relied on psychological assessments of stress, often using self-reported instruments such as the Perceived Stress Scale (PSS) and the Depression, Anxiety, and Stress Scales (DASS), among others [34]. More recently, physiological measures—especially RMSSD [6]—have been used to reflect stress during intervention-induced conditions [29–31]. For example, product purchase situations are often shaped by cognitive, affective, and sensorial factors, and considering all these, RMSSD can be used as an index predicting consumers’ behavior [6].

An extensive literature review by Kakaria et al. (2022) reveals that in state-of-the-art stress and HRV marketing research, portable devices are as commonly used and as accurate as non-portable devices. Moreover, stress and HRV measurements have explored diverse topics, including the influence of (ambient) noise on purchase intentions [35], the role of arousal in congruity-based product evaluation [36], consumer arousal and time pressure in online auctions [37], and consumer-brand connections [38]. The effects of stress

have also been examined more broadly, including its impact on performance, competitiveness, and the gender gap in the labor market [39]. Despite these advances, current marketing research has largely overlooked the role of physiological stress measurements in online shopping contexts, particularly in scenarios incorporating gamification elements.

4 Research model

Our research consists of two studies (Figure 1). The first study analyzes the effects of perceived gamification and stress on webstore retention, thereby contributing to a deeper understanding of how gamified design elements interact with user stress to influence continued use of online retail platforms. The second study assesses the effects of perceived gamification and stress on purchase intentions, thereby extending understanding of how gamified elements and stress responses influence consumers’ decision-making in online shopping contexts. For the physiological stress measurement, an arm strap sensor (electrical activity of the heart, ECG) was applied together with a visit to a selected webstore and survey questions. To assess perceived gamification features, we employed a six-item scale developed by De Canio et al. (2021). The scale captures the extent to which participants experience elements characteristic of gamification, including: the sensation of playing a game, completing tasks, feeling like a winner, collecting credits akin to those in a game, experiencing a gamified environment, and perceiving a challenging experience. For webstore retention we used webstore selection (participants were asked to select the most familiar and usual webstore for their clothing purchases) combined with the shop familiarity scale by Kent and Allen (1994), and for purchase intentions, the purchase intention scale by Dodds et. al. (1991).

In our model, we incorporated control variables guided by prior research. Specifically, gender was included, given its established influence on physiological indicators such as heart rate and stress responses [40]. In addition, nationality, reflecting participants’ cultural background, was included due to evidence that cultural factors can affect stress-related measures [41]. Finally, environmental literacy has been identified as a significant predictor of behavior, especially within our study context—fashion consumption [42].

5 Methodology

5.1 Study 1

Data for Study 1 was collected in Germany. After removing participants with excessive noise in heart rate recordings, participant counts were 74. Of the total accepted sample, 64.4% were women, and the median annual income range was £20,000–£29,999. In terms of occupation, 53% of respondents were full time students, 44% worked full-time and 3.4% were unemployed. For the physiological stress measurement, we adopted the Polar Verity Sense sensor [43], which is an arm-strap ECG sensor.

Study 1 began with a five-minute resting phase to establish baseline HRV, after which participants were asked to choose one of four retail websites—Patagonia, H&M, Shein, or Ralph Lauren—that they were most familiar with and usually consumed clothing from. After the store selection, participants completed a brief survey. Study 1 applied SD2/SD1 measured before the intervention, indicating

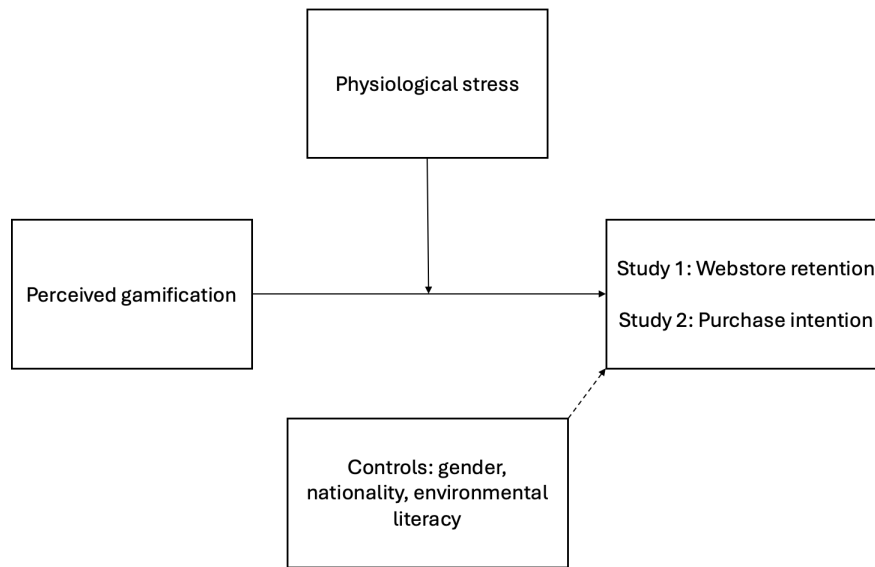


Figure 1: Research model.

at that time the long-term stress experienced by the participants. HRV data were processed using Kubios software.

5.2 Study 2

Data for study 2 was collected in Finland and accounted 55 participants after removing excessive noise in heart rate recordings. Of the total accepted sample, 65% were women. The median annual income range was € 20,000–€ 29,999. In terms of occupation, 25% of respondents were students, 20 % were retired, 18% worked as homemakers and 15% were full-time employed.

Study 2 also began with a five-minute resting phase to establish baseline HRV. Website selection was limited to Patagonia and Shein, representing the two extremes of gamification effects based on Study 1 findings. After that, participants were asked to spend at least five minutes in the selected website environments with no time limit. After the shopping task, participants completed a brief survey about their perceptions and experiences. HRV was monitored throughout both the baseline and shopping experience. Study 2 applied RMSSD, using the difference between baseline and during-experience stress (Geovanini et al., 2020), indicating short-term parasympathetic activity associated with relaxation, attention, and emotional regulation, hence more short-term stress [29, 30]. However, because RMSSD can also reflect long-term stress effects [31], it was considered a moderator in the research model.

6 Results

6.1 Study 1

For study 1 we conducted a regression analysis using SPSS Process Macro model 1 (Hayes, 2022). Results (Table 1) show that perceived gamification ($\beta = 0.9016$) and stress ($\beta = 0.8680$) positively affect retention to a familiar webstore, however, these effects were not

significant ($p = 0.0580$, $p = 0.1358$). Instead, the interaction of perceived gamification and stress was significant ($p = 0.0489$) and had a negative effect on retention ($\beta = -0.5131$).

6.2 Study 2

6.3 Post hoc analysis

In addition to analyzing the models for the composite gamification measurement, in both cases (Study 1 and 2) we ran the models with individual question items regarding the perceived gamification features (Figure 2). This explorative post hoc analysis showed that gamification elements making participants feel engaged and challenged, combined with stress, had the strongest negative effect on webstore retention. Regarding purchase intentions, collecting rewards as well as giving the feeling of a game-like experience combined with stress both had negative effects. On the other hand, a sense of achievement combined with stress had a positive effect on purchase intentions.

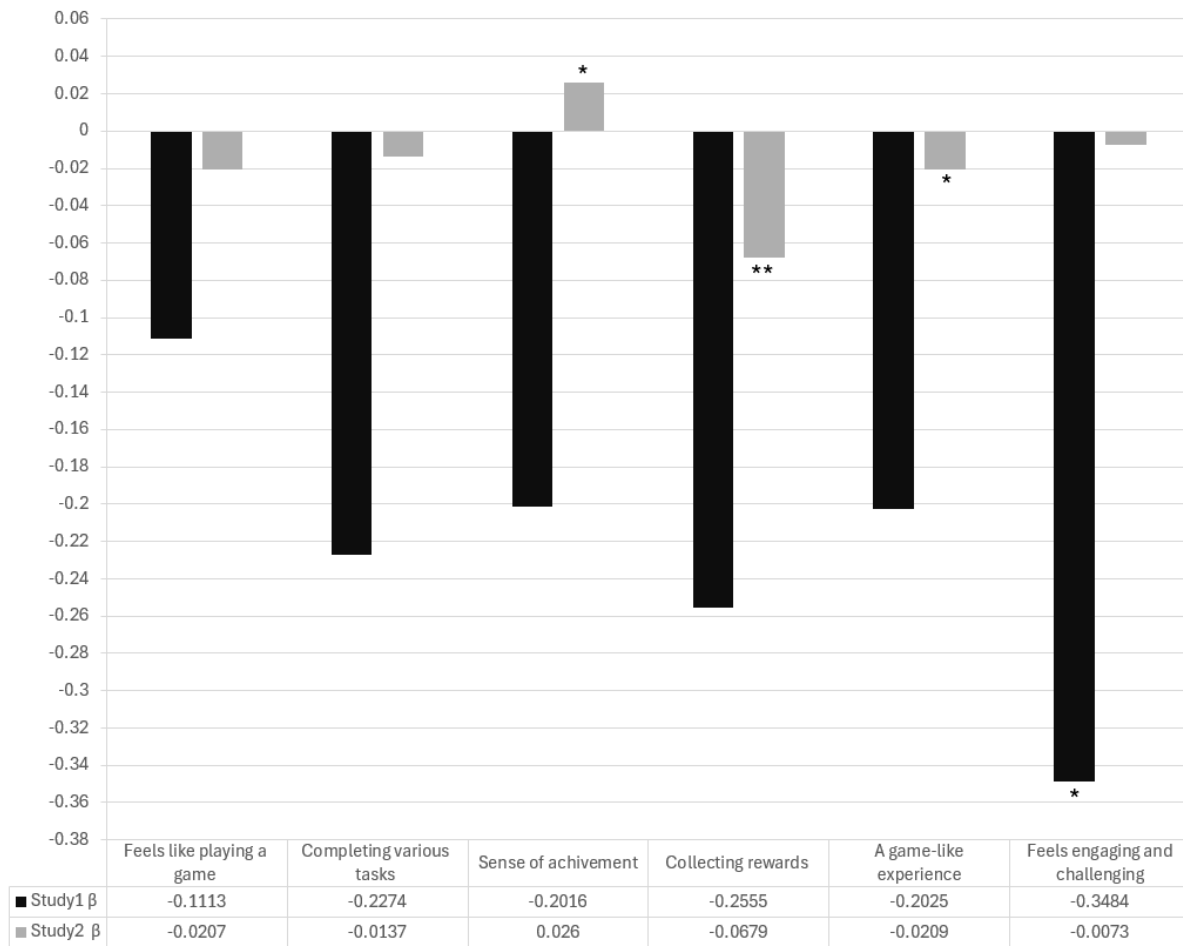
7 Discussion

In terms of theoretical contributions, while gamification has been found to positively affect webstore retention [43], our results show that under higher stress, consumers are less likely to do so. This finding aligns with previous research suggesting that under certain conditions, stress can lead to more impulsive buying, including webstore selection and thus decreased retention [44]. This pattern was also present in our second study, as the interaction of perceived gamification and stress led to a decrease in purchase intentions.

Furthermore, our findings suggest that consumers experiencing long-term stress show lower webstore retention because they are more likely avoiding challenging environments. In addition, collecting rewards and creating game-like experiences are not optimal gamification mechanisms for stressed consumers in increasing their purchase behavior. On the other hand, creating a sense of

Table 1: Study 1 and 2 results.

Variables	Study 1		Variables	Study 2	
Dependent variable	β	p	Dependent variable	β	p
Webstore retention			Purchase Intention		
Independent variable			Independent variable		
Perceived gamification	0.9016	0.0580	Perceived gamification	0.3596	0.2210
Moderators			Moderators		
Physiological stress (SD2/SD1)	0.8680	0.1358	Physiological stress (RMSSD)	0.0746	0.0676
Interaction term			Interaction term		
Perceived gamification x Physiological stress (SD2/SD1)	-0.5131	0.0489	Perceived gamification x Physiological stress (RMSSD)	-0.0428	0.0464
Control variables			Control variables		
Gender	0.2579	0.3168	Gender	0.0810	0.6421
Nationality	0.0088	0.6741	Nationality	-0.0664	0.0843
Environmental literacy	0.3015	0.0430	Environmental literacy	-0.6295	0.0013
Reliability			Reliability		
R ²	0.1478		R ²	0.3081	

**Figure 2: Regression coefficients (β) and significances of interactions of individual gamification mechanisms and physiological stress in studies 1 and 2. * $p < 0.10$, ** $p < 0.05$**

achievement experiences encouraged stressed consumers to make purchases.

These findings offer important implications for webstore designers and marketers developing gamification. First, these practitioners should ensure less stress among their customers through customer education so that they are less stressed when making e-commerce webstore selections. Moreover, while incorporating gamification features in their webstores, they should avoid increasing stress. More specifically, we recommend avoid creating challenging environments, reward systems, and game-like experiences for stressed consumers, while enabling achievement feelings instead. This study opens a new avenue in the field of stress and gamification research, calling for future research to validate these results with further experiments and contexts.

8 Limitations and future research

In this study the relatively small sample size used should be taken into consideration. Where possible, future studies should include larger samples to enhance the robustness and generalizability of the findings. Furthermore, future research should account more for cultural differences across countries and aim to employ research designs in which all participants are exposed to the same standardized gamified or non-gamified webstores.

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